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Gaia-Argus Administrator Guide

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Audience: System administrators responsible for deployment, configuration, and data storage management

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1. Application File Locations

1.1 Installer Output

The build process produces two Windows installers in:

```
src-tauri\target\release\bundle\
  msi\Gaia-Argus_0.1.0_x64_en-US.msi   ← WiX installer (preferred for enterprise deployment)
  nsis\Gaia-Argus_0.1.0_x64-setup.exe  ← NSIS installer (standalone)
```

Both installers place the application binary at the default system location for the chosen installer type. The `.exe` itself has no hard-coded paths; all runtime paths are resolved dynamically at launch.

1.2 Application Data Directory

Gaia-Argus stores all runtime data (SQLite historian, logs, cached state) in the Tauri application-data directory. The bundle identifier is `com.gaiacode.argus`, which determines the path:

OS	Path
Windows 10/11	`%APPDATA%\com.gaiacode.argus\`

Full expansion example:

```
C:\Users\<username>\AppData\Roaming\com.gaiacode.argus\
    gaiasignal_events.db       ← SQLite historian (events, picks, CCF stacks, Δv/v history)
```

This directory is created automatically on first launch if it does not exist. No manual setup is required.

1.3 Station Configuration Storage

Station configurations are stored in the browser-profile `localStorage` of the embedded Tauri WebView, not in a flat file on disk. The relevant key is `gaia_station_list`. The active station ID is stored under `gaia_active_station`.

Practical implication for administrators: Station configuration is **per-Windows-user-profile**. If the application is deployed for multiple Windows accounts on the same machine, each account maintains its own station list. To replicate a configuration across accounts, use the export/import procedure described in Section 2.5.

---

2. Station Configuration

2.1 Opening the Configuration Dialog

Click the **gear icon** (⚙) or the **Station Config** button in the top navigation bar. The Station Configuration modal opens as an overlay on the current view.

2.2 Field Reference

Each row in the dialog represents one station. The fields map directly to the internal `Station` TypeScript interface:

UI Label	Internal Field	Type	Description
Station Name	`name`	`string`	Human-readable display label. Appears in all dropdowns, tooltips, and the Sidebar station selector.
IP Address	`ip`	`string`	IPv4 address of the Tau digitizer (e.g. `192.168.1.68`). Used for the SeedLink TCP connection and for opening the datalogger web interface.
SEED IDs	`seed_ids`	`string`	Channel code selector string used when connecting SeedLink. See Section 3 for full syntax. Placeholder default: `XX.STA.00.H??`.
Lat	`lat`	`number`	WGS-84 latitude in decimal degrees (e.g. `39.9334`). Controls marker placement on the Network Map.
Lon	`lon`	`number`	WGS-84 longitude in decimal degrees (e.g. `32.8597`). Controls marker placement on the Network Map.
Elev (m)	`elevation`	`number`	Sensor elevation above mean sea level in metres. Used in HypoDD relocation geometry.
SeedLink Port	`port`	`number?`	TCP port of the SeedLink server on the Tau digitizer. **Default:** **`18000`**. Leave blank to use the default.
Web Port	`web_port`	`number?`	HTTP port of the Tau Web Interface embedded on the digitizer. **Default:** **`80`**. Set to `8080` or another value if the digitizer's web server is running on a non-standard port.

> **Precision note:** Latitude and longitude use a step of `0.0001` (≈ 11 m resolution at the equator) in the input field. For precise sensor placement, enter values to at least 4 decimal places (e.g. `39.9334`, not `39.93`).

2.3 Default Station List

On first launch with no saved configuration, the application loads 15 factory-default stations covering the default array layout:

```
STN-ORTAKOY, STN-01 through STN-13, STN-HAYDRLI
```

All default stations have empty IP, empty SEED IDs, and zeroed coordinates. They function as placeholders; configure each row before attempting a live connection.

2.4 Adding and Removing Stations

- **Add:** Click **"+ Add Station"** at the bottom of the station list. A new row is appended with a generated ID.
- **Remove:** Click the **Delete** button on the right of any row. This action is immediate and takes effect when the dialog is saved.

2.5 Export/Import Procedure

The station list is stored as JSON in `localStorage`. To export for backup or deployment replication:

1. Open the application on the source machine.
2. Open the browser DevTools console (press **F12** → Console tab, or right-click anywhere in the UI → Inspect).
3. Run:

```
copy(localStorage.getItem('gaia_station_list'))
```

4. The JSON is copied to the clipboard. Paste it into a `.json` file for safe-keeping.

To import on another machine or user account:

1. Open DevTools Console on the target machine.
2. Run:

```
localStorage.setItem('gaia_station_list', '<paste JSON here>');
location.reload();
```

3. The application reloads with the imported station list.

2.6 Station Configuration Schema (Reference)

The complete JSON schema for one station entry as stored in `localStorage`:

```
{
  "id":         "STN-01",
  "name":       "STN-01",
  "ip":         "192.168.1.68",
  "port":       18000,
  "web_port":   80,
  "seed_ids":   "XX.T743.00.H??",
  "lat":        39.9334,
  "lon":        32.8597,
  "elevation":  812.0
}
```

All fields except `id`, `name`, `ip`, and `seed_ids` are **optional**. The application handles missing values as follows:

Field	Missing value behaviour
`port`	Defaults to `18000` in the SeedLink connection dialog
`web_port`	Defaults to `80` when constructing the datalogger URL
`lat` / `lon`	Marker placed at `(0, 0)` (Gulf of Guinea) — always populate for map use
`elevation`	Treated as `0 m`; affects HypoDD depth geometry marginally

---

3. SeedLink Channel Codes & Wildcards

3.1 Code Format

The **SEED IDs** field and the **Channel Codes** input in the SeedLink Connection panel both accept a single `NET.STA.LOC.CHAN` selector string in four dot-separated fields:

```
XX.T743.00.H??
|   |    |  └── Channel code (3 chars: band + instrument + component)
|   |    └───── Location code (2 chars)
|   └────────── Station code
└──────────────── Network code
```

3.2 Wildcard Characters

Character	Meaning	Example	Matches
`?`	Exactly one arbitrary character	`H??`	`HHZ`, `HHN`, `HHE`, `HNZ`, `HLZ`, …
`*`	Zero or more arbitrary characters	`HH*`	`HHZ`, `HHN`, `HHE`
`*` (full field)	Any value for that field	`*.*.*.*`	Every stream (no filtering)

3.3 Wildcard Expansion for the Tau SeedLink Server

> **Important:** The GaiaCode Tau digitizer's SeedLink server **rejects SELECT commands containing `*` or `?` wildcard characters**. All wildcards are resolved on the client side before any SELECT commands are transmitted.

The `derive_selects()` function in `network.rs` performs this expansion automatically:

1. The `CHAN` field is inspected for `?` or `*`.
2. If a wildcard is found, the prefix before the first wildcard character is extracted (e.g. `H??` → prefix `H`).
3. A single-character prefix is extended to two characters by appending `N` (the instrument code for short-period sensors): `H` → `HN`.
4. Three explicit SELECT tokens are generated — one per component (Z, N, E):

Input `seed_ids`	Expanded SELECT tokens sent to server
`XX.T743.00.H??`	`00HNZ`, `00HNN`, `00HNE`
`XX.T743.00.HH?`	`00HHZ`, `00HHN`, `00HHE`
`XX.T743.00.HNZ`	`00HNZ` (exact match, no expansion)
`XX.T743.*.H??`	`00HNZ`, `00HNN`, `00HNE` (LOC `*` → default `00`)

The `LOC` field wildcard (`*` or empty) is resolved to `00` by default. If your installation uses a non-standard location code, specify it explicitly (e.g. `XX.T743.10.H??`).

3.4 Packet-Level Stream Filtering

After packets are received and decoded, a second filter (`stream_matches_filter`) is applied to each decoded SEED stream ID. This uses a standard two-character wildcard syntax:

- `?` matches exactly one character.

- `*` matches any sequence (including empty).
- An entirely empty `codes` string or `*.*.*.*` passes all streams through.

This filter ensures that even if the SeedLink server sends unrequested channels (some servers ignore selective subscriptions), only the configured channels are forwarded into the DSP pipeline.

3.5 Practical Examples

```
# Accept all high-gain short-period channels from station T743, network XX
XX.T743.00.H??

# Accept only the vertical component
XX.T743.00.HHZ

# Accept all channels from any station in network XX (broad filter)
XX.*.*.H??

# Accept every stream without filtering (useful for diagnostics)
*.*.*.*

# Accept a low-frequency broadband channel
XX.T743.00.BH?
```

---

4. SQLite Historian

4.1 Database File

```
%APPDATA%\com.gaiacode.argus\gaiasignal_events.db
```

The database is an ordinary SQLite 3 file. It can be opened with any SQLite client (DB Browser for SQLite, DBeaver, the `sqlite3` CLI) for inspection, export, or manual repair.

4.2 Schema

The schema is created automatically on first launch (`CREATE TABLE IF NOT EXISTS`). It is also migrated automatically — new columns are added non-destructively using `ALTER TABLE … ADD COLUMN` if they are absent from an older database file.

`events`

Column	Type	Description
`id`	`INTEGER PK AUTOINCREMENT`	Internal event identifier
`timestamp`	`REAL`	Unix epoch (seconds) of the estimated origin time
`latitude`	`REAL`	Degrees North
`longitude`	`REAL`	Degrees East
`depth`	`REAL`	Kilometres below surface
`magnitude`	`REAL`	Moment magnitude or local magnitude estimate

`phase_picks`

Column	Type	Description
`id`	`INTEGER PK`	Internal pick identifier
`event_id`	`INTEGER FK → events.id`	Parent event (cascade-deleted with event)
`station`	`TEXT`	Station code
`channel`	`TEXT`	Channel code (e.g. `HHZ`)
`phase_type`	`TEXT`	`"P"` or `"S"`
`timestamp`	`REAL`	Unix epoch of the arrival time
`amplitude`	`REAL`	Peak-to-peak amplitude at the pick
`polarity`	`TEXT`	First-motion polarity (`"U"`, `"D"`, or `""`) — added by migration
`is_manual`	`INTEGER`	`1` if operator-refined; `0` if automated — added by migration

`waveform_files`

Column	Type	Description
`id`	`INTEGER PK`	Internal identifier
`event_id`	`INTEGER FK → events.id`	Parent event
`file_path`	`TEXT`	Absolute path to the associated miniSEED waveform file

`ccf_stacks` (created by the Ambient Noise module)

Stores the rolling cross-correlation function stacks for each station pair. Rows are upserted by `process_noise_block`.

`dvv_history` (created by the Ambient Noise module)

Stores per-pair Δv/v time series. Each row represents one MWCS computation result.

4.3 Database Migrations

The `init_db` function applies migrations at startup. Currently implemented:

Migration	Condition	Action
Add `polarity` to `phase_picks`	Column absent	`ALTER TABLE phase_picks ADD COLUMN polarity TEXT`
Add `is_manual` to `phase_picks`	Column absent	`ALTER TABLE phase_picks ADD COLUMN is_manual INTEGER NOT NULL DEFAULT 0`

Migrations are idempotent — running them on an already-migrated database has no effect. No manual database version tracking is required.

4.4 Connection Pool

The application holds up to **5 concurrent SQLite connections** via `sqlx::SqlitePoolOptions`. For single-machine deployments this is more than sufficient. The database file supports WAL (Write-Ahead Logging) mode, which is SQLite's recommended mode for concurrent readers/writers.

---

5. Omega Directory Watcher

5.1 Purpose

The Omega Directory Watcher allows Gaia-Argus to index and display historical miniSEED files stored on the local filesystem in the standard Omega archiver directory layout, without requiring a running SeedLink server.

5.2 Expected Directory Structure

The watcher expects the standard Omega layout:

```
<BaseDir>\
    YYYY\
        YYYYMM\
            YYYYMMDD\
                <NETWORK>.<STATION>.<LOCATION>.<CHANNEL>.D.<YYYYDOY> .mseed
```

Example:

```
D:\GaiaData\
    2026\
        202607\
            20260730\
                XX.T743.00.HHZ.D.2026211.mseed
                XX.T743.00.HHN.D.2026211.mseed
                XX.T743.00.HHE.D.2026211.mseed
```

The scanner is **not strict about this layout** — it recursively descends any subdirectory structure and indexes every file with a `.mseed` extension (case-insensitive). Files in non-standard subdirectory layouts will still be indexed.

5.3 Starting the Watcher

1. In the **Data Ingestion** panel, expand the **Local Omega Directory** section.
2. Click **"Select Directory"** and browse to the Omega archive root (e.g. `D:\GaiaData`).
3. Click **"Watch Directory"**.
4. The backend Rust process (`watch_local_directory`) performs two actions in sequence:
   - **Initial scan:** Recursively reads all existing `.mseed` files in the tree. Each file is decoded with `gaia_dsp::miniseed::parse_mseed_bytes` (header-only; sample data is not loaded into

memory during the scan).

- **Live watch:** A `notify` filesystem watcher monitors the directory for new or modified `.mseed` files. Any file added to the tree after the initial scan is automatically indexed.

5.4 The Omega Catalog Table

As each file is indexed, an `omega_catalog_entry` IPC event is emitted and appended to the catalog table in the UI. The table displays:

Column	Description
SEED ID	Full `NET.STA.LOC.CHAN` stream identifier parsed from the miniSEED record header
Start	Start time of the record (local timezone)
End	End time of the record (local timezone)
Samples	Total sample count (`npts`)
Action	**"Plot Time Window"** button

The placeholder text above the table confirms the scan path:

```
Index-only scan of <watchPath>/YYYY/YYYYMM/YYYYMMDD/*.mseed.
No historical waveforms are plotted until a catalog row is selected.
```

5.5 Loading a Waveform from the Catalog

1. Locate the desired file in the catalog table.
2. Click **"Plot Time Window"** in the Action column.
3. `load_omega_file` is invoked; the full sample array is decoded and placed into the `currentWaveformData` store.
4. The **Canvas Waveform** renderer displays the trace immediately. All dependent modules (Helicorder, Response Spectra, Polarization) also gain access to the loaded data.

5.6 Memory Behaviour

The initial directory scan is **header-only** — only the miniSEED FIXED_SECTION_DATA_HEADER is parsed per file to extract stream ID, sample rate, start time, and sample count. The full sample data is **not** loaded into memory during the scan. This means a large archive (thousands of files) can be indexed without significant RAM usage. Sample data is loaded only when the operator explicitly clicks **"Plot Time Window"**.

5.7 Clearing the Catalog

Navigate away from the Omega section and back, or reload the application. The `watchCatalog` state is reset to `[]` on component mount.

---

6. Network Firewall Requirements

For a standard deployment with one Tau digitizer, the following inbound/outbound rules must be permitted on the operator workstation's Windows Firewall:

Direction	Protocol	Port	Service
Outbound	TCP	`18000` (default)	SeedLink connection to Tau digitizer
Outbound	TCP	`80` or `web_port`	Tau Web Interface (HTTP)
Inbound	—	—	No inbound ports required by Gaia-Argus

For deployments where the Tau digitizer and the operator PC are on separate VLANs, ensure the firewall policy permits TCP traffic from the PC's IP to the digitizer's IP on port `18000`. SeedLink is a standard TCP client-initiated protocol; no listening socket is opened on the PC side.

---

7. First-Time Deployment Checklist

- ☐ Install using .msi or NSIS installer on Windows 10/11 x64.
- ☐ Launch the application once to create the AppData directory and initialise the SQLite database.
- ☐ Open Station Configuration (gear icon).
  - ☐ Set Station Name for each array element.
  - ☐ Set IP Address to the digitizer's LAN IP (e.g. 192.168.1.68).
  - ☐ Set SEED IDs to the correct network/station/location/channel code (e.g. XX.T743.00.H??).
  - ☐ Set Lat/Lon/Elevation from the site survey GPS readings.
  - ☐ Set SeedLink Port (default 18000 — only change if Tau is non-standard).
  - ☐ Set Web Port (default 80 — only change if Tau uses a non-standard port).
  - ☐ Save configuration.
- ☐ Verify network reachability:
  - ☐ Ping digitizer IP from the workstation.
  - ☐ Confirm TCP port 18000 is reachable (test with: telnet <IP> 18000).
- ☐ Open Data Ingestion → SeedLink Connection.
  - ☐ Confirm IP, port, and SEED ID codes are populated from the station config.
  - ☐ Click Connect.
  - ☐ Confirm status bar shows "miniSEED listener started".
- ☐ Navigate to QC Matrix.
  - ☐ Confirm stream rows are present with green latency pills.
  - ☐ Confirm gap count is 0 on all channels.
- ☐ Navigate to Network Map.
  - ☐ Confirm station markers appear at correct geographic positions.
- ☐ Export station configuration (Section 2.5) and store the JSON alongside this document for disaster recovery.

8. Backup & Recovery

8.1 What to Back Up

Item	Location	Frequency
SQLite historian	`%APPDATA%\com.gaiacode.argus\gaiasignal_events.db`	Daily
Station configuration	`localStorage` JSON export (see Section 2.5)	After any config change
Omega miniSEED archive	`<BaseDir>\YYYY\…`	Continuous (managed by archiver)

8.2 SQLite Backup

SQLite supports safe online backup using the SQLite Online Backup API. The simplest method from the command line:

```
sqlite3 "%APPDATA%\com.gaiacode.argus\gaiasignal_events.db" ".backup backup.db"
```

Alternatively, copy the `.db` file while the application is **not** running, or while no write transaction is active. The WAL mode makes it safe to copy the file while the application is running, as long as you also copy the accompanying `-wal` and `-shm` files if they exist.

8.3 Restoring the Database

1. Stop the application.
2. Replace `gaiasignal_events.db` with the backup copy.
3. Delete any stale `-wal` and `-shm` files in the same directory.
4. Restart the application. The `init_db` migration runs automatically and will add any missing columns to the restored schema.

9. Troubleshooting

9.1 SeedLink Connection Fails

Symptom: Status bar shows `ERROR: TCP Connection failed` or connection times out after 5 seconds.

Checks:

1. Ping the digitizer IP from the workstation. If ping fails, the device is unreachable — check LAN cabling or switch port.
2. Confirm port `18000` is open: open a command prompt and run `telnet <IP> 18000`. A blank screen with no error indicates the port is open.
3. Check that the SEED IDs field is not empty. An empty `codes` string still attempts the connection but sends no SELECT commands, which may result in an empty stream on some server implementations.
4. Enable the **Debug Console** toggle in the SeedLink Connection section. The console log shows the raw SeedLink handshake exchange (`HELLO`, `STATION`, `SELECT`, `END`), which identifies exactly where the session is failing.

9.2 Map Markers Appear at (0°, 0°)

Symptom: Station markers are clustered in the Gulf of Guinea rather than the expected geographic location.

Cause: `lat`/`lon` fields in Station Configuration are zero (the factory default).

Fix: Open Station Configuration, enter the correct decimal-degree coordinates for each station, and save. The Network Map updates immediately on the next telemetry packet.

9.3 QC Matrix Shows No Rows

Symptom: The QC Matrix table is empty after connecting the SeedLink stream.

Cause: The `codes` filter in SEED IDs does not match any incoming stream IDs, or no packets have been received yet.

Fix: Verify that the `seed_ids` selector matches the stream IDs being sent by the digitizer. Enable the Debug Console in Data Ingestion to see raw decoded stream IDs as packets arrive. Use `*.*.*.*` temporarily as the `codes` value to disable filtering and confirm packets are arriving.

9.4 Omega Scan Does Not Populate the Catalog

Symptom: After clicking "Watch Directory", the catalog table remains empty.

Checks:

1. Confirm the selected directory contains `.mseed` files (any subdirectory depth).
2. Confirm the files are valid miniSEED format (not corrupted or truncated). Invalid files emit a `local_stream_error` event — enable the debug console to see error messages.
3. Check that the Windows user running the application has **read permission** on the selected directory and all subdirectories.

9.5 Database Migration Warning on Launch

Symptom: The application logs a warning about a missing column during startup.

Behaviour: This is expected when opening a database created by an older build. The migration system adds the missing column automatically with a safe default value. No data is lost. The warning will not appear again after the first successful launch with the new build.

9.6 High Latency on QC Matrix

Symptom: Latency pills are amber (≥ 1000 ms) persistently.

Checks:

1. Check that the workstation and digitizer clocks are synchronised (both should be NTP-synchronised). A large clock offset between client and server inflates the measured round-trip latency.
2. Check the LAN switch for port errors (`show interfaces` on a managed switch).
3. Verify the digitizer is not buffering due to a full ring buffer — check the Tau Web Interface diagnostics page.
4. If latency is consistently high but stable (not increasing), it may reflect genuine network propagation delay (e.g. a WAN link). This does not affect DSP correctness unless the delay exceeds the SeedLink buffer capacity.

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Gaia-Argus Operator Manual

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1. Getting Started

1.1 Launching the Application

Double-click `gaia-station-app.exe`. The Tauri shell opens and the Svelte UI initialises within the embedded WebView. No installation of additional runtimes is required — all dependencies are statically linked into the binary.

Application data (SQLite historian, configuration, logs) is stored at:

```
%APPDATA%\gaia-station-app\
```

1.2 Connecting a Live SeedLink Stream

1. Open the **Data Ingestion** panel from the sidebar.
2. Expand the **SeedLink Connection** section.
3. Enter the datalogger's **IP address** and **port** (default `18000`).
4. Enter the **network**, **station**, and **selector** codes (e.g. `XX`, `T743`, `00H??`).
5. Click **Connect**. The status bar updates to confirm the stream is live.

Once connected, all real-time modules (Network Map, Helicorder, QC Matrix, Polarization) update automatically as packets arrive.

1.3 Station Configuration

Select the active station from the **Station Selector** at the top of the sidebar. The active station controls which stream is used by the Response Spectra, Phase Picker, and Polarization modules. Multiple stations appear in the Network Map simultaneously regardless of the active selection.

---

2. Navigation & Layout

The application uses a persistent left sidebar for module navigation. Click any entry to switch the main view:

Sidebar Entry	Module
Network Map	Real-time station map with PGA markers and trigger vectors
QC Matrix	SeedLink stream health, latency, and gap monitoring
Helicorder	24-hour continuous drum plot
Response Spectra	FFT/PSD snapshot workspace
Event Catalog	Historical event list, phase picking, HypoDD relocation
Polarization	3-component particle motion & azimuth compass
Ambient Noise	Cross-correlation Δv/v structural health dashboard
Site Char.	MASW dispersion image and Vs30 classification
Data Ingestion	SeedLink, offline miniSEED, Omega directory loader

Right-click any sidebar entry for quick-access context menu actions (where supported).

---

3. Network Map

The Network Map provides a live geographic overview of all configured stations and their current seismic status.

3.1 Station Markers

Each station is represented by a **circular SVG marker** on the Leaflet map. The fill colour encodes the most recent Peak Ground Acceleration (PGA) reading on a continuous scale:

Colour	Approximate PGA	Interpretation
Teal (`#0EADB2`)	< 0.02 g	Below detection threshold — quiet background noise
Transitioning to amber/red	0.02 g – 0.10 g	Elevated ground motion; monitor closely
Red (`#ef4444`)	≥ 0.10 g	Maximum displayed scale — strong motion event

Markers with active triggers **pulse** with a CSS animation to draw immediate attention. A tooltip is available by hovering (or right-clicking on touch displays) showing station ID, real-time PGA, latency, and network code.

> **Note:** Markers only pulse and change colour if PGA exceeds the detection threshold (0.02 g). Quiet stations display in teal regardless of noise floor.

3.2 Back-Azimuth Trigger Vectors

When a **Polarization trigger** fires for a station (rectilinearity R ≥ the configured threshold), a **polyline vector** is drawn on the map extending outward from that station's marker in the computed back-azimuth direction.

- The vector length is proportional to the trigger intensity (configurable, default ≈ 10 km map-scale).
- The vector is **removed automatically** when the trigger condition clears.
- Multiple simultaneous vectors from different stations allow rapid visual triangulation of the apparent source direction.

3.3 Interacting with the Map

- **Pan:** Click and drag.
- **Zoom:** Mouse wheel, or the `+` / `−` controls.
- **Fly to station:** Click a station marker; the map animates (`flyTo`) to centre on that station and slightly increases zoom.

- **Station selection:** Clicking a marker also sets that station as the **active station** for all dependent modules.

---

4. QC Matrix

The QC Matrix is the primary quality control surface for monitoring SeedLink stream health across all configured channels.

4.1 Reading the Matrix

Each row represents one stream (identified by its full SEED channel code, e.g. `XX.T743.00.HHZ`). The columns are:

Column	Meaning
Stream ID	Full network.station.location.channel code
Status pill	`Active` (teal) or `Muted` (amber)
Latency pill	Round-trip latency in milliseconds. Green `< 1000 ms`; amber `≥ 1000 ms`
Gaps	Count of detected sample gaps since last reset. `0` shown as a quiet teal pill; any non-zero count displays as a red badge
Mute toggle	Checkbox switch to temporarily suppress this stream

4.2 Latency Interpretation

- **Green (< 1000 ms):** Stream is arriving within normal bounds for a local SeedLink server.
- **Amber (≥ 1000 ms):** Elevated latency. Causes include network congestion, a slow datalogger, or the TCP buffer filling up. If latency climbs steadily, consider restarting the SeedLink connection from the Data Ingestion panel.

4.3 Gap Counter

A gap is recorded each time the backend detects a discontinuity in the sample sequence (a missing or out-of-order packet). Gaps degrade DSP quality for all modules consuming that stream.

- A **red gap badge** (e.g. `3`) indicates the number of gap events since the counter was last cleared.
- A gap count of `0` displays as a quiet teal `0` pill.

If gaps persist on a specific channel, suspect a hardware issue with the datalogger ADC board or a network routing problem between the digitizer and the host PC.

4.4 Muting a Malfunctioning Stream

If a channel is producing noise, saturated signals, or continuous gaps that are corrupting array-wide analyses, it can be **temporarily muted** without disconnecting:

1. Locate the row for the affected stream.
2. Click the **toggle switch** in the rightmost column.
3. The status pill changes from `Active` to `Muted`.
4. The stream continues to be received and logged, but it is excluded from DSP computations (polarization, F-K beamforming, ambient noise stacking) until unmuted.

To **unmute**, click the toggle again. The status pill returns to `Active` and the stream re-enters all processing pipelines from the next incoming packet.

> **Important:** Muting a stream does not stop data archiving. All raw samples continue to be written to the SQLite historian regardless of mute state.

---

5. Helicorder

The Helicorder provides a continuous 24-hour drum-plot view — the standard seismological display for rapid visual scanning of a day's seismic activity.

5.1 Understanding the Display

The plot is divided into **horizontal time rows**, each representing a fixed duration (e.g. 30 minutes). Time advances left to right within each row, then wraps to the beginning of the next row below. The most recent data is always at the **bottom of the display**.

The Y-axis of each row is the normalised waveform amplitude. Gaia-Argus computes a **global peak amplitude** (`globalPeak`) across *all* rows and locks every row to the same `[−yScale, +yScale]` range (`yScale = globalPeak × 1.2`). This ensures that a large-amplitude event in one row does not compress the scale of adjacent rows, which is the standard professional display convention for spotting both high-energy and micro-seismic events on the same screen.

5.2 Live Mode

When a SeedLink stream is active, new samples are appended to the current row in real time. The active row advances automatically when the row duration elapses. No operator action is required.

The row label on the left shows the UTC start time of that row.

5.3 Offline miniSEED Mode

The Helicorder can also display pre-recorded data:

1. Navigate to **Data Ingestion** and load a `.mseed` file using **Select Data File** or drag-and-drop into the drop zone.
2. Return to the **Helicorder** view.
3. The ingested data populates the rows in chronological order. Row labels reflect the UTC timestamps from the miniSEED record headers.

This mode is particularly useful for reviewing historical events on the day of the occurrence without needing a live connection.

5.4 Identifying Events

On the drum plot, look for:

- **Sharp vertical deflections** on a row — the classic impulsive P-wave arrival signature.
- **Sustained higher-amplitude oscillations** following the initial deflection — surface waves or S-wave coda.
- **Uniform low-amplitude trace** — quiet background noise or a muted stream.
- **Flat zero-line** on an otherwise active display — a data gap; check the QC Matrix for that stream.

To investigate any feature, note its row label (start time) and minute position within the row, then use the **Event Catalog** or **Phase Picker** to access the precise waveform data for that time window.

---

6. Response Spectra

The Response Spectra module generates FFT power spectral density (PSD) plots and response spectra from the current waveform data. It is a **snapshot-only** workspace: all computation is triggered manually to prevent the heavy FFT from blocking the live ingestion thread.

6.1 The Snapshot Model

The module deliberately does **not** update automatically with each incoming packet. This design choice prevents the FFT computation (which is O(n log n) and runs synchronously with the Tauri UI) from causing stuttering in the live waveform display, Helicorder, or QC Matrix. The most recent waveform data is held in a background store; the Response Spectra view reads from that store only when you explicitly trigger a capture.

6.2 Capturing a Snapshot

1. Ensure live data is streaming **or** that an offline file has been loaded via Data Ingestion.
2. Navigate to the **Response Spectra** view.
3. Optionally adjust the channel selector and frequency-band inputs if shown.
4. Click the **"Capture Snapshot & Plot"** button (teal tonal button, top of the panel).
5. The button label changes to **"Computing…"** while the backend processes the FFT. This typically completes within 1–2 seconds.
6. The PSD plot and response spectra curves render in the workspace. Live data continues to flow into the background store undisturbed.

6.3 Taking Multiple Snapshots

Each click of **"Capture Snapshot & Plot"** replaces the current spectral display with a fresh snapshot from the latest available data. To compare two time periods:

1. Capture the first snapshot.
2. Use your operating system's **Print to PDF** or a screen-capture tool to save the first result.
3. Wait for the desired time or load a different file.
4. Click **"Capture Snapshot & Plot"** again.

6.4 Interpreting the PSD

- **Flat broadband PSD** across 1–20 Hz — typical quiet background noise floor.
- **Sharp spectral peak at a single frequency** — potential resonance of a structural element or instrument artefact.
- **Elevated energy in the 1–5 Hz band** — P-wave or S-wave energy from a nearby event.
- **Extremely high spectral amplitude at all frequencies** — sensor saturation or cable fault; correlate with QC Matrix gap count.

---

7. Event Catalog & Phase Picker

The Event Catalog is the primary post-event analysis workspace. It displays the historical event list stored in the SQLite historian and provides interactive tools for refining phase picks and preparing inputs for the HypoDD relocation solver.

7.1 The Event List

The catalog table lists all detected events with columns for event ID, origin time (UTC), latitude, longitude, depth, and magnitude estimate. Events are sorted by most-recent first by default.

Selecting events for analysis:

- Click the **checkbox** on the left of any row to select it.
- Use the **Select All** checkbox in the table header to select the entire catalog.
- Selected events are highlighted and their IDs are queued for HypoDD relocation (see Section 8).

7.2 Opening the Phase Picker

Click on any event row (not the checkbox, but anywhere else in the row) to open the **Interactive Phase Picker** for that event.

The Phase Picker opens as a **3-panel Plotly chart**, one panel per seismic component (Z, N, E), with:

- The waveform trace for each component plotted on a UTC time axis.
- **Vertical marker lines** representing the automated phase picks stored for that event:
  - **Solid lines** = P-wave picks (automated or manual).
  - **Dotted lines** = S-wave picks (automated or manual).
  - **Teal / green lines** = manual picks (refined by an operator).
  - **Red lines** = automated picks (not yet manually reviewed).

Channel colour coding: Z-component in teal, N-component in amber, E-component in indigo.

7.3 Refining Phase Arrival Times

The chart is rendered with `edits: { shapePosition: true }` enabled. This means every pick marker is **draggable** directly on the chart.

To adjust a P or S arrival:

1. Hover over a pick line until the cursor changes to a horizontal resize arrow.
2. Click and **drag the line** left or right to the correct arrival time.
3. Release the mouse. A `plotly_relayout` event fires internally.
4. The updated timestamp is passed immediately to the parent component via the `onPickChange` callback, which writes the revised pick to the SQLite database.

No save button is required — the pick update is committed automatically on drag release.

> **Tip:** Use the chart's built-in **pan mode** (the hand cursor, default `dragmode: "pan"`) to scroll along the time axis before fine-tuning a pick. Scroll the mouse wheel to zoom in on the arrival for precise placement.

7.4 Verifying Pick Quality

After refinement, assess pick quality visually:

- A well-placed **P pick** sits at the onset of the first impulsive deflection on the Z-component, before the waveform amplitude significantly exceeds the noise floor.
- A well-placed **S pick** sits at the onset of the larger-amplitude shear arrival, typically most visible on the N and E components.
- If picks appear on the wrong component panel, check the channel code mapping in the station configuration.

---

8. HypoDD Relocation

The HypoDD double-difference relocation solver refines hypocenter coordinates for a group of events using differential travel times. It requires **at least two events** with associated phase picks.

8.1 Selecting Events

1. In the Event Catalog, tick the checkboxes for the events to be relocated.
2. A minimum of **2 events** must be selected. The relocation quality improves with more events sharing common station observations.
3. The selected event count is displayed in the status bar at the bottom of the catalog.

8.2 Running the Solver

1. With events selected, click the **"Run HypoDD"** button (teal tonal button in the catalog toolbar).
2. The button label changes to **"Relocating…"** and the row is locked.
3. The Rust backend runs the iterative SVD-based double-difference inversion:
   - 1-D velocity model: Vp = 6.0 km/s, Vs = 3.47 km/s.
   - Convergence threshold: residual norm $\|\Delta \mathbf{m}\|_2 < 0.1$ m.
4. On completion the status bar shows:

   ```
   HypoDD complete — N iterations, RMS X.XXXX s
   ```

   where `N` is the iteration count and the RMS is the final differential travel-time residual.

8.3 Viewing Relocation Results

After the solver completes, a **3D Plotly modal** opens automatically:

- **Faded markers** (reduced opacity) show the **original** hypocenter positions.
- **Solid markers** show the **relocated** positions.
- **Thin lines** connect each original–relocated pair to visualise the shift vector.

- Axes are geographic: X = Longitude (°E), Y = Latitude (°N), Z = Depth (km, downward positive).

Interact with the 3D plot:

- **Rotate:** Left-click drag.
- **Zoom:** Mouse wheel.
- **Pan:** Right-click drag.
- **Reset view:** Double-click.

Close the modal to return to the catalog. The relocated coordinates are stored back in the SQLite `events` table and will persist across application restarts.

8.4 Interpreting the Output

- **Small shift vectors** (< 1 km) with low RMS (< 0.3 s) indicate the automated locations were already well-constrained.
- **Large consistent directional shifts** across all events suggest a systematic station timing error or velocity model mismatch.
- **Diverging residuals** (RMS increasing across iterations) indicate insufficient phase picks or poorly distributed station geometry for the event cluster.

---

9. Polarization Analysis

The Polarization module analyses the three-component particle motion of the active station to determine P-wave incidence angle, back-azimuth, and rectilinearity.

9.1 Controls

Control	Description
Window size (s)	Duration of each analysis window (0.5 – 4 s). Shorter windows track rapid onsets; longer windows are more stable on low-SNR signals.
Overlap (0–0.99)	Fractional overlap between successive windows. Higher overlap gives smoother time resolution at the cost of more computation.
Rectilinearity threshold	Minimum R value (0.0 – 1.0) for a window to be reported as a P-wave trigger. Default 0.75.
Channel selector	Select which Z/N/E channel group to analyse if multiple 3-C sets are present.

9.2 Running an Analysis

1. Ensure waveform data is available (live stream or loaded file).
2. Set the window and threshold parameters.
3. Click **"Re-analyse"**. The Rust backend processes all windows synchronously and emits one `polarization_telemetry` IPC event per qualifying window.

9.3 Reading the Metric Cards

- **Rectilinearity (R):** Ranges from 0.0 (perfectly circular motion = noise) to 1.0 (perfectly linear = pure P-wave). Values above the configured threshold trigger the `⚡ P-WAVE TRIGGER` pulse badge.
- **Azimuth (°):** Back-azimuth of the incoming wavefront, measured clockwise from North. A value of 270° means the apparent source is to the west.
- **Incidence (°):** Angle of the P-wave ray from the vertical. 0° = vertically incident; 90° = horizontally propagating.
- **Eigenvalues (λ_1, λ_2, λ_3):** Sorted descending. The ratio $\lambda_1/(\lambda_1+\lambda_2+\lambda_3)$ quantifies how much of the motion energy is concentrated in the principal direction.

9.4 Hodogram

The **3D scatter hodogram** plots the particle trajectory in [E, N, Z] space for the most recent analysis window. The **red line** through the cloud represents the principal eigenvector direction — this is the vector from which the back-azimuth and incidence are derived. A tightly clustered, elongated cloud aligned with the red line confirms a high-rectilinearity P-wave arrival.

9.5 Azimuth Compass

The polar plot shows the back-azimuth history:

- **Faded teal markers:** Previous window results (history trail).
- **Solid spoke:** Current azimuth. Turns red during a P-wave trigger.
- **Dashed spoke:** Reciprocal direction (azimuth ± 180°), indicating the ±180° ambiguity.

When multiple stations are visible on the **Network Map** simultaneously with their azimuth vectors drawn, the intersection of vectors from at least two stations provides a rapid visual cross-bearing to the apparent source.

---

10. Ambient Noise (Δv/v)

The Ambient Noise module monitors long-term structural velocity changes by cross-correlating continuous background noise between station pairs. A sustained decrease in Δv/v may indicate material degradation, micro-fracturing, or ground saturation before visible failure.

10.1 Station Pair Selection

1. Use the two **Station** dropdown menus to select **Station A** and **Station B**.
2. The cross-correlation is always computed as A × B (the pair key is order-independent; `A-B` and `B-A` retrieve the same stored stack).

10.2 Processing Controls

Field	Description
Sample Rate (Hz)	Sampling rate of the input data (must match the actual channel rate).
Min / Max freq (Hz)	Passband for pre-correlation bandpass filtering.
MWCS Window (s)	Duration of each Moving Window Cross-Spectral analysis sub-window.
MWCS Step (s)	Step between successive MWCS windows.
History Days	Number of days of Δv/v history to display on the timeline.

10.3 Loading a One-Hour Block

1. Click **"Process 1-Hour Block"**.
2. The backend applies 1-bit temporal normalisation and spectral whitening to both channels, computes the cross-correlation function (CCF), and adds the result to the rolling stack in the SQLite database.
3. The **Stack: N records** badge in the station pair header updates to reflect the new count.

Repeat this process for each 1-hour block to build up the reference Green's function over 24 hours or more. The first 24 records (hours) are recommended before Δv/v values are considered reliable.

10.4 Computing Δv/v

1. Click **"Compute Δv/v"** after accumulating at least a few stack records.
2. The MWCS algorithm computes the differential time delay δt between the current CCF and the reference stack in sliding sub-windows.
3. The result appears in the **Δv/v Timeline** and is stored in the `dvv_history` SQLite table.

10.5 Alert Threshold

1. Enter a threshold value in the **Alert Threshold** field (e.g. `-1.0` for a 1% velocity decrease).
2. If any Δv/v history value is equal to or more negative than the threshold, the **red alert banner** appears at the top of the module.
3. Rows in the history table that breach the threshold are highlighted in the error colour.

A sustained Δv/v below −1% warrants a site inspection. Values below −3% should be treated as a structural health alert requiring immediate engineering assessment.

10.6 CCF Daily Stack Plot

The lower chart displays the reference cross-correlation waveform:

- **Dashed teal trace** (0.4 opacity): The reference Green's function (multi-hour/day stack).
- **Solid teal trace:** The most recently processed 1-hour CCF.

The X-axis is the lag time (seconds). The dominant coherent arrival(s) represent the empirical Green's function between the two stations — typically surface waves whose group velocity can be estimated from the lag and inter-station distance.

---

11. Site Characterization (MASW)

The Site Characterization module performs Multi-channel Analysis of Surface Waves (MASW) to measure shear-wave velocity structure and compute the NEHRP Vs30 site classification.

11.1 Generating the Dispersion Image

1. Ensure waveform data from a linear array shot or ambient noise record is loaded.
2. Confirm the **Sample Rate** and **Receiver Spacing** inputs are correct for your array geometry.
3. Click **"Compute MASW Image"**. The slant-stack p-τ transform runs in the Rust backend (`spawn_blocking`) and returns the dispersion energy image as a heatmap.
4. The image displays frequency on the X-axis (Hz) and phase velocity on the Y-axis (m/s), with colour intensity showing the stacked energy.

The fundamental mode dispersion curve appears as the brightest ridge in the image.

11.2 Picking the Dispersion Curve

1. After the image renders, **click directly on the brightest ridge** at each frequency.
2. Each click adds a pick point (displayed as a white dot overlay).
3. Continue picking across the full frequency band visible in the image.

4. Picks are listed in the **Pick Table** on the right with their frequency and phase velocity values.

To remove a pick, click its row in the Pick Table and use the delete control.

11.3 Inverting for the Vs Profile

1. Once a sufficient dispersion curve has been picked (minimum 5–10 points spanning at least one octave), click **"Invert Picks"**.
2. The damped least-squares inversion iterates over 128 velocity bins (100–1500 m/s) to fit a layered-earth Vs profile to the picked curve.
3. The resulting **Vs depth profile** is plotted as a step function on the right panel.

11.4 Vs30 and NEHRP Classification

The module automatically computes **Vs30** — the harmonic-mean shear-wave velocity over the uppermost 30 m — using the EN 1998-1 formula and displays the NEHRP site class badge:

Badge	Vs30	Site Class
A/B	≥ 760 m/s	Rock
C	360 – 759 m/s	Very dense soil / soft rock
D	180 – 359 m/s	Stiff soil
E	< 180 m/s	Soft soil — high amplification risk

The NEHRP class directly informs foundation design codes and should be reported to the structural engineer if it differs from the design-basis assumption.

---

12. Data Ingestion

12.1 Single miniSEED File

1. Click **"Select Data File"** in the Stored Files section.
2. Browse to and select a `.mseed` file.
3. The file is loaded via the binary IPC pipeline (`load_waveform_binary`): the Rust backend packs `f32` amplitude samples into a raw byte buffer and returns it as a `Uint8Array` — no JSON serialization occurs.
4. The **Canvas Waveform** renderer displays the loaded trace immediately. The shared waveform store is also populated so the Helicorder, Response Spectra, and Polarization modules can access the same data.

12.2 3-Component File Set

1. Click **"Select 3-Component Files"** and select three files (Z, N, E channels).
2. Each file is loaded via the same binary pipeline and displayed as a separate Canvas Waveform row.
3. The Polarization module can now be triggered against the loaded 3-C data.

12.3 Omega Directory

1. Use **"Watch Directory"** to point Gaia-Argus at a local Omega directory.
2. The directory watcher scans for new files continuously. Newly detected files are listed in the **Watched Files** catalog.
3. Click any catalog entry to load it immediately into the Canvas Waveform display.

12.4 Tau Datalogger Web Interface

The Tau digitizer runs an embedded HTTP server. Click the **"Open Tau Web Interface"** link (or navigate directly to `http://<digitizer-ip>`) within the application's browser pane to access the digitizer's channel configuration, gain settings, RTCC synchronisation controls, and diagnostic registers — without leaving the Gaia-Argus window.

Appendix A — Status Quick-Reference

Element	State	Colour
Station marker	Quiet (< 0.02 g PGA)	Teal `#0EADB2`
Station marker	Strong motion	Red `#ef4444`
Latency pill	Good (< 1000 ms)	Green `#10b981`
Latency pill	Elevated (≥ 1000 ms)	Amber `#f59e0b`
Gap badge	Non-zero	Red badge
QC status pill	Active stream	Teal
QC status pill	Muted stream	Amber
Phase pick line	Automated	Red
Phase pick line	Manual (refined)	Green
P pick	Arrival type	Solid line
S pick	Arrival type	Dotted line
Polarization badge	R ≥ threshold	⚡ pulse — error-red border
Δv/v alert banner	Value ≤ threshold	Red error-container background
NEHRP badge	See Vs30 table above	—

Appendix B — Keyboard & Mouse Reference

Action	Input
Zoom waveform (Canvas)	Mouse wheel
Pan waveform (Canvas)	Left-click drag
Reset waveform view	Double-click
Touch zoom (Canvas)	Two-finger pinch
Touch pan (Canvas)	Single-finger drag
Drag phase pick	Left-click drag on pick line
Pan Phase Picker	Left-click drag on chart
Rotate HypoDD 3D plot	Left-click drag
Zoom HypoDD 3D plot	Mouse wheel
Pan HypoDD 3D plot	Right-click drag
Reset 3D view	Double-click
Open station tooltip (Map)	Hover or right-click marker
Fly to station (Map)	Left-click marker

13. Operational Modal Analysis (OMA / FDD)

The OMA module uses Frequency Domain Decomposition to identify the natural frequencies and damping ratios of a monitored structure from ambient vibration alone — no artificial excitation required. It is the primary tool for bridge span resonance surveys, floor vibration assessments, and railway infrastructure structural health benchmarking.

13.1 Theory in Brief

FDD decomposes the **Cross-Spectral Density (CSD) matrix** of the multi-channel response at each frequency line using Singular Value Decomposition. Near a structural resonance, the CSD matrix becomes rank-1 and the first singular value exhibits a sharp local maximum. The corresponding left singular vector is the **mode shape** — the relative amplitude and phase of each sensor at that frequency.

Signal path (Rust backend):

1. `csd_matrix()` — Welch-style estimation with a **Hann window**, `nperseg` samples per block, **50% overlap** (`noverlap = nperseg / 2`), and standard PSD normalisation (`1 / (Fs × N)`). The result is a Hermitian `n_channels × n_channels` complex CSD matrix per frequency bin.
2. `fdd_svd()` — nalgebra `SVD::new()` applied to the **magnitude** of each CSD matrix. Returns up to 3 singular values per frequency bin (`n_singular = min(n_channels, 3)`).
3. `find_peak_indices()` — local maxima detection with prominence gating on SV1.
4. `half_power_damping()` — half-power bandwidth method: ζ = (f_upper − f_lower) / (2 × f_n), where f_upper and f_lower are the frequencies where SV1 drops to its peak value / √2.

13.2 Data Requirements

- **Minimum 2 channels.** The module enforces this (`"FDD requires at least two channels"`).
- All selected channels **must have identical sample counts**. If channels were loaded from separate files with differing durations, trim to the shortest common window before FDD — or reload as a synchronous 3-component file set.
- For structural applications, channels should represent **spatially distributed sensors** at different measurement points (e.g. accelerometers on different floors, bridge spans, or rail sleeper positions).
- A minimum record length of roughly **10 × (1/f_min)** seconds is recommended. For a structure with a 1 Hz fundamental mode, aim for at least 10 seconds; for sub-Hz modes (long-span bridges) use longer records.

13.3 Setting Up the Array

1. Load a synchronous multi-channel file (or active live stream) via **Data Ingestion**. All streams must share the same sample rate and start time.
2. Navigate to **OMA / FDD** from the Sidebar.
3. The **channel checkbox list** at the top of the module populates automatically from the loaded streams. Each checkbox label shows the full SEED stream ID.
4. By default, **all available streams are selected**. De-select any noisy or malfunctioning channel identified in the QC Matrix.

> **Tip:** For a 3-component sensor on one structural element (e.g. Z, N, E at the bridge mid-span), select all three. For a distributed array (Z-only sensors at multiple locations), select only the relevant Z channels. FDD treats each checked stream as one row of the response matrix; the geometry is implicit in the sensor placement.

13.4 Computation Controls

Control	Options	Description
FFT block size (nperseg)	256, 512, 1024, 2048	Samples per Welch averaging block. Controls the frequency resolution: Δf = Fs / nperseg. Larger values give finer frequency resolution but fewer averaged blocks, increasing variance. **Default: 1024.**
Decimation	1×, 2×, 4×, 8×	Integer downsampling applied before FDD. Use to reduce computation time on very long records or to focus on low-frequency structural modes. Effective Fs = Fs_original / decimation.

Choosing nperseg:

- Structures with modes above 5 Hz → `512` or `256` (wide bins acceptable; fast).
- Structures with modes in 1–5 Hz → `1024` (default, good balance).
- Long-span bridges or base-isolated structures (< 1 Hz) → `2048` at full sample rate, or `1024` with 2× decimation.

The minimum auto-selected nperseg is `64`; if the provided value is less than 64 or zero, the backend substitutes `min(1024, N/4)` where N is the record length.

13.5 Reading the FDD Plot

The FDD plot shows **frequency (Hz) on a logarithmic X-axis** vs **singular value amplitude on the Y-axis**, with up to three singular value curves:

Curve	Colour	Meaning
SV 1 (first/dominant)	Teal `#0EADB2`	Captures the dominant response direction at each frequency. Peaks identify structural modes.
SV 2	Amber `#F59E0B`	Second orthogonal direction. Peaks here that align with SV1 peaks indicate closely-spaced modes or multi-directional response.
SV 3	Indigo `#6366F1`	Third direction. Generally remains flat unless the structure has three or more identifiable modes.

Identifying modal peaks:

1. Scan SV1 for distinct, isolated peaks.
2. Each peak is a candidate natural frequency. The algorithm auto-detects peaks with a minimum prominence of `0.1 × mean(SV1)`.

3. Peaks that appear on **both SV1 and SV2** may indicate repeated modes (e.g. symmetric two-span bridge with near-identical bending frequencies in two axes).
4. Peaks in SV1 that **do not appear** in SV2/SV3 are likely well-separated single-axis modes.

> **Warning:** Harmonics of machinery vibration (fans, motors, traffic loading) appear as sharp, narrow peaks. Structural modes tend to have slightly broader bandwidth. A structural mode will also shift in frequency during temperature changes or after damage; a machinery harmonic will not.

13.6 Modal Properties Table

Below the FDD plot, a table lists every automatically detected mode:

Column	Description
Mode	Sequential mode number (1 = lowest detected frequency)
Natural Frequency f_n (Hz)	Frequency of the SV1 peak, reported to 3 decimal places
Damping Ratio ζ	Half-power bandwidth estimate, reported as a percentage

Interpreting damping ratios:

- Reinforced concrete structures: typically 2–5%.
- Steel structures: typically 0.5–2%.
- Base-isolated structures: 5–20% (isolation system contributes).
- Values below 0.5% or above 15% should be treated with caution — they may reflect a poorly resolved half-power bandwidth rather than a physical property.

13.7 Tracking Structural Change

To use OMA for structural health monitoring over time:

1. Establish a **baseline** survey immediately after installation, under known environmental conditions (ambient temperature noted).
2. Record the modal frequencies and damping ratios from the table.
3. Repeat the FDD analysis at regular intervals (weekly or after significant loading events).
4. A **progressive downward shift in f_n** (> 3–5%) on any mode indicates stiffness degradation. Combined with a corresponding **increase in ζ** (energy dissipation increasing, often indicating cracking or joint loosening), this warrants a detailed structural inspection.

---

14. HVSR — Horizontal-to-Vertical Spectral Ratio

The HVSR module implements Nakamura's (1989) technique for measuring the **fundamental resonance frequency f_0** of the soil column at a site. It requires only a single three-component ambient noise recording and is the standard method for rapid seismic site characterisation before or alongside the MASW survey.

14.1 Theory

The H/V ratio is defined as:

```
H/V(f) = √[(PSD_N(f) + PSD_E(f)) / 2] / PSD_Z(f)
```

where PSD_N, PSD_E, and PSD_Z are the power spectral density estimates of the North, East, and Vertical components respectively. A peak in the H/V curve at frequency f_0 indicates the resonance of a soft soil layer overlying a stiffer substrate.

Backend processing chain (`compute_hvsr` in `gaia_dsp`):

1. Hann-windowed FFT of each component.
2. **Konno-Ohmachi smoothing** applied to each amplitude spectrum with bandwidth parameter `b = 40.0` — the SESAME (2004) recommended value for HVSR.
3. H/V ratio computed from the smoothed spectra.
4. `f0` reported as the frequency of the maximum H/V amplitude above 0.5 Hz.

14.2 Recording Requirements

For a reliable HVSR measurement:

- **Record duration:** Minimum 20 minutes of uninterrupted ambient noise. Longer is better — 30–60 minutes is standard practice. The quality of the H/V estimate depends on the stationarity of the noise field; transient anthropogenic sources (vehicles, footsteps) during the recording window degrade the result.
- **Component orientation:** N and E channels must be genuinely horizontal (tilt < 2°). The Z channel must be genuinely vertical. Verify sensor levelling in the Tau Web Interface before recording.
- **Record type:** Load a pre-recorded ambient noise file via **Data Ingestion**. HVSR should **not** be computed on triggered event records — the strong P and S wave arrivals violate the stationarity assumption of the Nakamura technique.
- **File loading:** Use the **Select Data File** button in Data Ingestion and choose the 3-component miniSEED file for the site. Alternatively, load from an Omega directory catalog row.

14.3 Channel Assignment

1. Navigate to **HVSR** from the Sidebar.
2. Three dropdown selectors appear: **Z stream**, **N stream**, **E stream**.
3. On initial load, the module auto-assigns streams by matching channel codes ending in `Z`, `N`, and `E` respectively. If auto-assignment succeeds, streams are shown in the dropdowns and computation proceeds immediately.
4. If auto-assignment fails (e.g. unusual channel naming conventions), manually select the correct stream from each dropdown. The available options list all currently loaded stream IDs.

> **Important:** HVSR requires all three components to contain data. If any stream is empty, the error banner `"HVSR requires N, E, and Z streams."` is displayed. Reload the waveform data and verify all three components are present using the QC Matrix.

14.4 Reading the H/V Plot

The HVSR plot shows **frequency (Hz) on a logarithmic X-axis** vs **H/V amplitude (dimensionless) on a linear Y-axis**:

- **Teal curve:** The H/V ratio spectrum — the primary result.
- **Red dashed vertical line:** The automatically detected **f_0** value, labelled in the legend as `f0 = X.XX Hz`.
- The H/V amplitude at f_0 should ideally be ≥ 2.0 for a SESAME-compliant peak; values below 2.0 may indicate a weak impedance contrast or insufficient recording duration.

14.5 Identifying f_0

The **fundamental resonance frequency f_0** is reported in the plot legend as a decimal value in Hz. Interpretation:

f_0	Approximate Soil Type
> 10 Hz	Very shallow soft layer (< 5 m) or very stiff site (rock)
2–10 Hz	Moderate depth soft layer (5–30 m)
0.5–2 Hz	Deep soft sediment (30–200 m)
< 0.5 Hz	Very deep basin fill (> 200 m)

The corresponding approximate **soil column depth H** can be estimated from:

$$H \approx Vs / (4 \times f_0)$$

where Vs is the average shear-wave velocity of the soft layer (in m/s). Use the Vs30 value from the MASW module as a first approximation.

14.6 Reliability Assessment

A reliable HVSR peak must satisfy all three SESAME (2004) criteria:

1. The peak frequency f_0 > 10 / record_duration_seconds.
2. The H/V amplitude at f_0 is ≥ 2.0 (at least double the background).
3. The peak is reproduced consistently when the record is split into sub-windows.

Gaia-Argus reports f_0 from the full-record spectrum. Manual splitting and re-computing (by loading sub-segments via Data Ingestion) is recommended to verify stability. A f_0 that shifts by more than ±20% across sub-windows should be treated as unreliable.

---

15. Kinematics

The Kinematics module integrates 3-component acceleration waveforms into velocity and displacement time series, and provides calibration-aware unit conversion across the full physical unit chain. It is the standard workflow for computing Peak Ground Velocity (PGV) and Peak Ground Displacement (PGD) from recorded accelerations, and for assessing structural inter-storey drift from floor-mounted sensors.

15.1 Data Pipeline

Backend Rust processing (`integrate_kinematics` in `gaia_dsp`):

1. **DC removal** — `detrend_constant()` applied to the raw acceleration array. This removes the mean offset so that a residual DC bias on the ADC does not integrate into a ramping velocity signal.
2. **Velocity** — cumulative trapezoidal integration of the detrended acceleration. `detrend_linear()` applied to the result (OLS linear trend removal) to suppress long-period drift introduced by integration of baseline noise.
3. **Displacement** — cumulative trapezoidal integration of the velocity. `detrend_linear()` applied again.

All three stages are NaN-safe.

Frontend calibration (`unitScaling.ts`):

Before invoking the Rust integration, raw integer ADC counts are converted to physical acceleration (m/s²) via `scaleCountsToUnit()` using the calibration constants from `calibrationStore`. The output of the Rust integrator is therefore in physical SI units: velocity in m/s, displacement in m.

15.2 Channel Selection

Three dropdowns at the bottom of the controls section select the Z, N, and E streams:

- On first load, streams are auto-assigned by channel code suffix (`Z`, `N`, `E`).
- If multiple location codes are present in the loaded data (e.g. `00` and `10` for two sensor levels in a building), changing the **Z stream** automatically locks N and E to the matching location code, preventing a cross-sensor mismatch.
- To analyse a single-level sensor suite, leave the auto-assignment as-is.

15.3 Metric Toggle

Three buttons at the top of the controls bar select the displayed quantity:

Button	Units available	Typical use
Acceleration	m/s², g, Gal, cm/s², in/s²	Direct ADC output in physical units; verify sensor calibration; compute PGA
Velocity	m/s, cm/s, mm/s, in/s	Compute PGV; rail vibration assessment (EN 16584, ISO 2631)
Displacement	m, cm, mm, in	Compute PGD; inter-storey drift for structural damage assessment

Switching metric re-selects an appropriate default unit automatically:

Metric	Default unit on switch
Acceleration	m/s²
Velocity	m/s
Displacement	m

The **Unit** dropdown lists only units compatible with the selected metric. Incompatible units are disabled (greyed out) in the list.

15.4 High-Pass Filter

Integration amplifies low-frequency noise: a small DC residual after detrending, or a low-frequency microseismic signal, will accumulate into a large displacement ramp that obscures the true structural response. The **high-pass (HP) filter** suppresses these low-frequency artefacts.

1. Enable the filter by toggling the **HP filter** checkbox in the analysis controls bar.
2. The **HP cutoff (Hz)** input becomes active. Enter the desired corner frequency.
3. The filter is applied **before** invoking the Rust integrator, on the physical acceleration data:

- A **4th-order Butterworth zero-phase filter** (`zero_phase_filter` in `gaia_dsp/filter.rs`) is applied via the `apply_filter` Tauri command with `filterType: "highpass"`.
- Zero-phase filtering (`filtfilt` — forward pass then time-reversed backward pass) introduces **no group delay** and preserves the phase of the acceleration onset.

4. The integration is then performed on the filtered acceleration.

Recommended HP cutoff values:

Application	Recommended cutoff
Building inter-storey drift (structural)	0.1 – 0.5 Hz
Rail vibration / short-period events	0.5 – 1.0 Hz
High-frequency machinery monitoring	1.0 – 5.0 Hz
Very long records with strong DC drift	0.05 Hz

Setting the cutoff too high will remove genuine low-frequency structural response. Setting it too low will not adequately suppress the integration drift. The default starting value is **0.1 Hz**.

> **Note:** There is no explicit low-pass filter control in this module. The physical bandwidth is already limited by the sensor's anti-alias filter (hardware). If a software low-pass is required (e.g. to remove high-frequency noise before displacement computation), use the Response Spectra module to verify the signal content first, then set an appropriate HP cutoff that is high enough to suppress drift but low enough to preserve the motion of interest.

15.5 Display Controls

Control	Description
Window (s)	The trailing time window displayed on the X-axis. Adjusts to show the most recent N seconds of the computed series. Default 60 s. Minimum 1 s. The computation always uses the full loaded record; this control only adjusts the plot view.
Auto Y toggle	When on (default), the Y-axis scales automatically to the data range in the current window. When off, **Y min** and **Y max** inputs become active for manual axis limits.
Y min / Y max	Manual Y-axis limits when Auto Y is off. Useful for comparing successive records on the same scale, or for detecting very small displacements against a fixed baseline.
View Mode: Overlaid / Stacked	**Overlaid:** All three components (Z in teal, N in amber, E in indigo) are drawn on a single shared Y-axis. Useful for comparing relative amplitudes between components. **Stacked:** Each component gets its own Y-axis sub-panel. Useful for inspecting individual channels when amplitudes differ greatly between components.

The display is capped at **6,000 samples** per component per render (`MAX_POINTS = 6000`) to maintain smooth canvas interaction. For long high-sample-rate records, reduce the **Window (s)** value to stay within this limit, or use the Decimation control in the OMA module to reduce the effective sample rate for structural analyses.

15.6 Interpreting the Results

Acceleration traces should show centred oscillations around zero with no persistent offset. A visible DC offset after HP filtering indicates a residual sensor tilt — check sensor levelling.

Velocity traces should return smoothly to near-zero after the wave train passes. A persistent non-zero velocity after the motion ends indicates HP cutoff is too low (the integration ramp is not fully suppressed). Increase the HP cutoff in 0.1 Hz increments until the post-motion velocity baseline is flat.

Displacement traces tell the same story at lower frequency. For seismic applications, the **peak displacement** (maximum excursion from zero) is the PGD. For structural applications (floor sensors), the displacement represents the **absolute floor displacement** relative to the pre-motion rest position. **Inter-storey drift ratio** = (displacement at floor N − displacement at floor N−1) / storey height — compute this by running Kinematics on two floor channels independently and subtracting.

---

16. Orbit 3D — Particle Motion Visualization

The Orbit 3D module renders the three-dimensional trajectory of ground motion as a continuous curve in [East, North, Vertical] displacement space. It is the primary tool for visually confirming wave type and propagation direction before running the quantitative Polarization analysis.

16.1 What It Shows

The module computes 3-C displacement by invoking the same `compute_kinematics` backend used by the Kinematics module (DC removal → trapezoidal integration → linear detrend, twice). It then plots the resulting displacement triplet as a single `scatter3d` **line trace** where:

- **X-axis:** East displacement (m)
- **Y-axis:** North displacement (m)
- **Z-axis:** Vertical displacement (m)
- **Line colour:** Encodes elapsed time via the **Viridis** colour scale (dark purple = beginning of record, bright yellow = end). This allows the temporal evolution of the trajectory to be read directly from the colour gradient.

Line width is fixed at 4 px. The scene `aspectmode` is set to `"data"`, meaning the axis scales reflect the actual physical displacement ratios — no artificial exaggeration.

16.2 Data Requirements

- Three-component (Z, N, E) waveform data must be loaded.
- All three channels must contain samples. If any is empty, the error `"3D orbit requires Z, N, and E streams."` appears.
- The module works identically on live-stream data and loaded files.

16.3 Channel Selection

Three dropdowns at the top of the module select the **Z stream**, **N stream**, and **E stream**. On first load, streams are auto-assigned by matching channel codes ending in `Z`, `N`, and `E`. Computation fires automatically when all three are populated.

To analyse a different sensor or location code, change the Z stream dropdown — the N and E dropdowns do not auto-lock to the same location code in this module (unlike Kinematics), so ensure all three are from the same physical sensor.

16.4 Windowing for Phase Analysis

The Orbit 3D module always plots the **full loaded record**. To isolate a specific phase arrival (e.g. examine only the P-wave particle motion before S-wave energy contaminates the trajectory):

1. Use **Data Ingestion → Select Data File** to load the full event record.
2. Navigate to **Helicorder** to identify the approximate P-wave arrival time and duration.
3. Return to **Data Ingestion** and use the **Watch Directory** catalog to identify the exact file. Note the start timestamp.
4. Load a **trimmed sub-segment** from the Omega catalog: select the row covering the desired 1–5 second P-wave window and click **"Plot Time Window"**. This replaces the waveform store with the trimmed data.
5. Navigate to **Orbit 3D**. The plot now shows only the P-wave particle trajectory.

What to expect from a P-wave window:

A clean P-wave arrival produces a **elongated, needle-like trajectory** aligned with the direction of propagation. The particle moves back and forth along a nearly linear path. The Viridis colouring will show a clear directional outward stroke (purple → yellow at maximum displacement) followed by the return stroke.

An S-wave window, by contrast, produces a **looping or helical trajectory** — motion is largely transverse (in the N-E plane for a horizontally propagating S-wave) with relatively little vertical component.

Ambient noise (no phase) produces a **roughly spherical cloud** with no preferred orientation.

16.5 Interacting with the 3D Plot

The Plotly `scatter3d` canvas supports full 3D interaction. All interactions are mouse-driven with no keyboard shortcuts required:

Interaction	Mouse Action
Rotate	Left-click drag anywhere on the scene
Zoom	Mouse wheel (scroll up = zoom in, scroll down = zoom out)
Pan	Right-click drag
Reset view	Double-click on the scene
Axis rescale	Click the **Autoscale** button (↔ icon) in the Plotly toolbar
Save image	Click the **camera icon** in the toolbar to export a PNG

Recommended views for interpretation:

- **Horizontal projection (view from above):** Rotate so the Z-axis points directly at you (looking straight down). The N-E plane is now a 2D map view. The elongation direction of the P-wave

trajectory gives the back-azimuth directly — the angle from north is the azimuth of the incoming wave.

- **Vertical plane:** Rotate to look along the E-axis (east face). The N-Z plane shows the incidence angle — a steeply inclined trajectory (close to vertical) indicates a nearly vertically incident wave (local/shallow source); a shallow trajectory indicates a distant or deep source.
- **Free 3D view:** The default perspective is the most informative starting point for assessing whether the trajectory is linear (P-wave), planar (SV-wave), or volumetric (noise/coda).

16.6 Relationship to Polarization Module

The Orbit 3D plot is a qualitative visual confirmation. For quantitative polarization parameters (rectilinearity R, back-azimuth in degrees, incidence angle), use the **Polarization** module (Section 9 / Section 17), which applies the eigendecomposition of the covariance matrix to the same 3-C data.

---

17. Polarization — Quantitative Wavefield Analysis

The Polarization module automates the extraction of P-wave arrival direction and wave quality from 3-component data using covariance matrix eigendecomposition — the mathematical foundation of the Vidale (1986) and Flinn (1965) polarization methods.

17.1 Theory: Covariance Eigendecomposition

For each analysis window of W samples, the algorithm constructs the **3×3 sample covariance matrix C** of the de-meaned [Z, N, E] displacement vector:

```
C[i,j] = (1/W) Σ (xi_k − x̄i)(xj_k − x̄j)    for i,j ∈ {Z, N, E}
```

`nalgebra::SymmetricEigen::new(C)` decomposes C into eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ and corresponding eigenvectors. The eigenvectors form an orthogonal coordinate frame of the particle motion ellipsoid:

- **Principal eigenvector (λ_1):** Axis of maximum variance — the direction of dominant particle motion.
- **λ_2 eigenvector:** Intermediate axis.
- **λ_3 eigenvector:** Minor axis.

Derived parameters (computed each window):

Parameter	Formula	Physical meaning
Rectilinearity R	`1 − (λ₂ + λ₃) / (2λ₁)`	0 = circular noise; 1 = perfect linear P-wave
Back-azimuth	`atan2(vE, vN)`, CW from N	Direction the wave came from
Incidence θ	`atan2(√(vN² + vE²),	vZ

The ±180° back-azimuth ambiguity is resolved by the sign of vZ: if vZ < 0, the azimuth is flipped by 180°, so that the reported azimuth always points toward the source.

17.2 Analysis Controls

Control	Options	Default	Description
Window (s)	0.5, 1.0, 2.0, 4.0 s	1.0 s	Duration of each sliding covariance window. Shorter windows track rapid P-wave onsets better; longer windows reduce variance on weak signals.
Overlap	25%, 50%, 75%	50%	Fractional overlap between successive windows. 75% overlap → step = 0.25 × window (densest sampling, highest computation).
Rect threshold	0.0 – 1.0, step 0.05	0.7	Minimum rectilinearity R for a window to be reported as a trigger. Windows below this threshold are computed but not emitted, unless no window in the entire record exceeds the threshold — in that case the window with the highest R is emitted as a fallback.
Channels	Z/N/E (3-D), Z/N	Z/N/E	Select the component subset for analysis.

17.3 Running an Analysis

1. Load waveform data (live stream or file).
2. Set window, overlap, and threshold parameters for the expected signal type.
3. Click **"Re-analyse"**. The button label changes to **"Analysing…"** while the Rust backend processes all windows synchronously.
4. On completion, the Polarization view updates immediately. If the Polarization module is not the active view, the `triggerState` Svelte store is updated silently and the NetworkMap draws or removes vectors without requiring user navigation.

17.4 Trigger Badge

When any analysis window has R ≥ `rectThreshold`, the ⚡ **P-WAVE TRIGGER** badge appears at the top of the module. The badge pulses with a 0.8-second CSS animation (`badge-pulse`) on each new qualifying trigger event. The badge is visible regardless of which sub-panel (hodogram, compass, metrics) is currently active.

The `triggerFlash` state lasts for 800 ms; the badge remains visible (without pulsing) until R drops below threshold in a subsequent Re-analyse call.

17.5 Automatic Network Map Integration

The Polarization module and the NetworkMap share data via the **`triggerState` Svelte store** (`Record<stationId, TriggerEvent>`). This store is a global singleton subscribed to by both modules simultaneously.

Data flow:

```
run_3c_polarization (Rust)
    → emit("polarization_telemetry", payload)
        → Polarization.svelte listen() handler
            → triggerState.update({ [stationId]: event })
                → NetworkMap.svelte triggerState.subscribe()
                    → drawVector(station, backAzimuth, VECTOR_KM=5.0)
```

On every `triggerState` change, the NetworkMap's reactive update block runs for every configured station:

- **Trigger active (backAzimuth present):** A `leaflet.Polyline` is drawn from the station marker's `[lat, lon]` to a point `VECTOR_KM = 5.0 km` away in the `backAzimuth` direction, using the Haversine destination formula. The polyline colour is red during the active `triggerFlash` window and teal otherwise. The vector length is fixed at 5 km map-scale regardless of zoom level (the endpoint is a geographic coordinate).
- **Trigger cleared:** The polyline is removed from the map and deleted from the `vectors` Map.

This means an operator monitoring the **Network Map view** during an event will see back-azimuth vectors appear automatically on every station that detects a P-wave — no navigation to the Polarization module is required.

17.6 History Table and Trend

The module retains the last **120 analysis windows** (`MAX_HISTORY = 120`) in the `history` array. These are displayed in the history table below the compass, with rows highlighted in the trigger colour when R ≥ threshold.

The compass **history trail** plots all 120 azimuths as faded teal scatter markers, allowing the operator to see whether the back-azimuth has been consistent over the analysis period (a tightly clustered trail indicates a stable source direction) or scattered (indicating noise or a rotating source).

---

18. Ambient Noise (Δv/v) — Structural Velocity Monitoring

The Ambient Noise module implements passive seismic interferometry to track long-term changes in the shear-wave velocity of a monitored structure or site. A sustained negative Δv/v trend is an early indicator of material degradation, micro-cracking, or stress redistribution in concrete, steel, or soil before any visible structural change occurs.

18.1 Physical Principle

Ambient seismic noise (traffic, wind, machinery, ocean microseisms) is continuously present at any monitoring site. Noise recorded simultaneously at two sensors contains **coherent wavefield energy** propagating between them. Cross-correlating these two noise records extracts the **empirical Green's function** — equivalent to the response one station would record if the other were an impulsive source.

If the medium's elastic properties change (e.g. a crack forms, a joint loosens, or the concrete water content shifts), seismic wave velocities change, and the Green's function **stretches or compresses in time**. The **Moving Window Cross-Spectral (MWCS)** analysis quantifies this as a relative velocity change:

```
Δv/v = −(δt / t)
```

where δt is the measured time delay between the reference and current Green's function at lag time t.

18.2 Station Pair Setup

1. Navigate to **Ambient Noise** from the Sidebar.
2. Select **Station A** and **Station B** from the two dropdown menus. Available options are drawn from the configured station list.
3. The pair is displayed as `A × B` in the header. The pair key is order-independent — `A × B` and `B × A` retrieve the same stored correlation stack.

4. For structural monitoring (e.g. a bridge), Station A and Station B should be sensors at different structural positions whose wave path passes through the monitored element (e.g. both deck sensors, or one deck and one abutment sensor).

18.3 Processing Parameters

Field	Default	Description
Sample Rate (Hz)	100 Hz	Must match the actual channel sample rate. A mismatch causes systematic Δv/v errors.
Min freq (Hz)	1.0 Hz	Lower corner of the MWCS analysis passband.
Max freq (Hz)	20.0 Hz	Upper corner of the MWCS analysis passband. Set to ≤ Nyquist/2 (50 Hz for 100 sps data).
MWCS Window (s)	5.0 s	Duration of each MWCS sub-window for phase-delay estimation.
MWCS Step (s)	1.0 s	Step between successive MWCS sub-windows.
History Days	30 days	Number of days of Δv/v history to display on the timeline.
Alert Threshold (%)	−1.0 %	Any Δv/v value ≤ this value activates the red alert banner.

18.4 Building the Reference Stack

The reference cross-correlation function (the "coda") is the cornerstone of Δv/v measurement. All subsequent Δv/v values are measured as deviations from this reference. A stable, well-averaged reference is essential for sensitivity.

Step-by-step procedure:

1. Set **Station A** and **Station B** to the desired monitoring pair.
2. Confirm **Sample Rate** matches the data.
3. Load the first 1-hour noise block via Data Ingestion.
4. Click **"Process 1-Hour Block"**. The backend runs:
   - **1-bit temporal normalisation** on each channel (suppresses earthquake transients and machinery bursts).
   - **Spectral whitening** (divides each frequency bin by its smoothed amplitude; flattens the noise spectrum so no single frequency dominates the correlation).
   - **FFT cross-correlation** of the two whitened, normalised channels.

- The resulting CCF is **added to the running stack** in the `ccf_stacks` SQLite table for this pair.

5. Repeat for each subsequent 1-hour block. The **Stack: N records** badge in the pair header increments with each successful block.
6. The **CCF Daily Stack Plot** updates after each block:
   - The **dashed teal trace** (0.4 opacity) shows the reference Green's function accumulated so far.
   - After adding new data, the **solid teal trace** shows the most recently computed CCF.

How many blocks are needed?

The reference stack quality improves with the number of averaged blocks. For routine structural monitoring:

- **Minimum viable reference:** 24 blocks (one per hour over 24 hours). The Green's function is typically stable after this.
- **Recommended reference:** 7 × 24 = 168 blocks (one full week). Averages out diurnal temperature cycles and traffic variations.
- **High-precision monitoring:** 30+ days. Required for detecting Δv/v changes smaller than 0.1%.

The reference stack is stored in SQLite and survives application restarts. There is no need to reprocess historical blocks unless the station geometry or equipment changes.

18.5 Computing Δv/v

Once a reference stack has been accumulated:

1. Load a fresh noise block (the "current" measurement window) via Data Ingestion.
2. Click **"Compute Δv/v"**. The backend: a. Processes the current block (1-bit norm + whitening + CCF) — the same pipeline as **Process 1-Hour Block** but without adding to the stack. b. Runs the **MWCS algorithm** between the current CCF and the reference stack: slides the `winS`-second sub-window across both CCFs in `stepS` steps, measures the phase delay in each frequency bin via cross-spectral phase slope, and fits a weighted least-squares line to δt vs. t to extract the single scalar Δv/v. c. Appends the result to the `dvv_history` SQLite table with the current epoch.
3. The status bar updates: `Δv/v = X.XXX % ± Y.YYY % (N windows)`.
4. The **Δv/v Timeline** plot updates with the new data point.

18.6 Reading the Δv/v Timeline

The timeline shows **UTC date/time on the X-axis** and **Δv/v (%) on the Y-axis**, with:

- **Teal data points** connected by a line — each point is one `compute_dvv` call.
- **Error bars** — ± uncertainty from the MWCS weighted least-squares fit.
- **Red dashed horizontal line** — the alert threshold level.

- **Red highlighted rows** in the history table below the plot — any measurement at or below the threshold.

Interpreting the timeline:

Observation	Likely interpretation
Stable Δv/v near 0% over weeks	Healthy structure; no significant velocity change
Seasonal sinusoidal oscillation (±0.1–0.5%)	Thermo-elastic effect (temperature-driven velocity) — normal
Progressive monotonic decrease (weeks to months)	Accumulating damage, micro-cracking, or progressive saturation
Sudden step decrease coinciding with a seismic event	Co-seismic damage or dynamic triggering of existing weakness
Recovery after a decrease (Δv/v returning toward 0%)	Healing — common in soils after rainfall saturation; less common in concrete

18.7 Alert Threshold Management

1. Set the **Alert Threshold** field to the velocity decrease you consider a structural health alert (e.g. `−1.0` for 1% decrease).
2. The alert activates when any history entry satisfies `dvvPercent ≤ threshold`.
3. The **red alert banner** appears at the top of the module when active.
4. If the most recent Δv/v value breaches the threshold, the banner text reads: *"⚠ Velocity change alert: Δv/v = X.X%"*

A sustained breach warrants correlating with:

- Temperature logs (rule out thermo-elastic seasonal effect).
- Recent precipitation or groundwater data (for soil-structure systems).
- Event catalog (check for recent earthquakes or heavy loading events near the site).
- Physical inspection of the monitored element.

18.8 Operational Cadence Recommendations

Monitoring type	Recommended cadence
Post-earthquake rapid assessment	Process 1 block immediately; compute Δv/v within 30 minutes
Active construction monitoring	Daily — one block per shift
Long-term SHM baseline	Weekly — one block per week during stable periods
Pre-failure alert system	Continuous — automate via the Omega Directory Watcher

For continuous automated monitoring, configure the **Omega Directory Watcher** (Admin Guide §5) to index new miniSEED files as they arrive from the Tau digitizer. Each new 1-hour file appearing in the Omega tree triggers an `omega_catalog_entry` event. Load it from the catalog, process the block, compute Δv/v, and observe the timeline for threshold breaches. This workflow requires no manual operator intervention once the watcher is running.

Appendix C — Module Quick-Reference

Module	Sidebar Entry	Min. Data Requirement	Key Output
OMA / FDD	OMA / FDD	≥ 2 channels, synchronous, equal length	Modal frequencies f_n (Hz) + damping ratio ζ (%)
HVSR	HVSR	3-C ambient noise, Z + N + E	Fundamental frequency f_0 (Hz)
Kinematics	Kinematics	3-C waveform, Z + N + E	Velocity & displacement time series, PGV, PGD
MASW	Site Char.	Multi-channel surface wave record	Vs30 (m/s) + NEHRP class
Polarization	Polarization	3-C stream, Z + N + E	Rectilinearity R, back-azimuth, incidence angle
Ambient Noise	Ambient Noise	2 stations, continuous archive	Δv/v (%) vs time

19. Live Waveform — Real-Time Telemetry Viewer

The Live Waveform module is the primary continuous monitoring display in Gaia-Argus. It renders incoming SeedLink telemetry and loaded miniSEED data as a scrolling time-series chart, and doubles as the manual phase-picking and amplitude measurement interface for event analysis.

19.1 Rendering Architecture

The viewer uses **Plotly `scatter` traces** (SVG-backed) with a hard render cap of **`MAX_SAMPLES = 4,000` points per trace**. Incoming data from the `liveIntake.ts` listener appends to the `currentWaveformData` Svelte store continuously; the display renders from that store at a **capped rate of ~10 FPS** (100 ms `setInterval`), decoupled from the data intake rate. This architecture means:

- No live packets are ever dropped by rendering — the intake listener and the render timer are completely independent.
- RAM usage is bounded: regardless of how long a SeedLink session runs, the chart never holds more than 4,000 points per visible channel.
- The **Pause button** freezes the display only — the backend continues writing to the store while paused. On resume, the latest `MAX_SAMPLES` points are shown immediately.

Timestamps are stored as **Unix seconds** throughout the Rust backend, SQLite, and all IPC invocations. They are converted to ISO-8601 date strings only at the Plotly boundary (`secToIso()`), avoiding the Plotly `scattergl` float32 precision artefacts that caused the diagonal-line rendering bug in earlier WebGL builds.

19.2 Channel Selector

Three checkboxes at the top of the control bar select which channels to display:

Checkbox	Channel suffix	Trace colour
Z	Vertical	Teal `#0EADB2`
N	North	Amber `#F59E0B`
E	East	Indigo `#6366F1`

Any combination may be active simultaneously. Unchecking a channel removes it from the render immediately without disrupting the intake for the other channels. If all three are checked and **View Mode: Stacked** is selected, each channel occupies its own horizontal lane with an independent Y-axis. **View Mode: Overlaid** plots all channels on the same Y-axis.

The **Analysis stream** dropdown selects which single channel feeds the STA/LTA detector, phase-pick saving, and amplitude measurement. This selection is independent of the channel checkboxes — a channel can be analysed without being displayed, and vice versa.

19.3 Mouse Interaction: Pan, Zoom, and Selection

The chart uses Plotly's standard date-axis interaction model. No mode buttons need to be toggled for basic navigation:

Action	Mouse Gesture
Zoom in	Click and drag to draw a zoom rectangle (default drag mode)
Pan	After zooming in, double-click to reset; or use the Plotly toolbar Pan button (hand icon)
Reset to full view	Double-click anywhere on the chart
Zoom on a single axis	Drag along the X-axis label (time zoom only) or Y-axis label (amplitude zoom only)
Zoom with mouse wheel	Scroll over the chart (Plotly default)

> **Note on drag mode:** When **Box Select Mode** is checked, the default drag gesture switches from zoom to amplitude-measurement selection (see §19.7). Uncheck it to restore zoom behaviour.

The time window displayed is governed by the **Window (s)** control (default 40 s, minimum 1 s). The chart always anchors to the **most recent sample** — it is a trailing window, not a fixed-start view. To inspect earlier data, pause the stream first, then use Plotly zoom/pan to navigate backward in the buffered record.

19.4 Time Window Control

The **Window (s)** number input (in the second control row) sets the trailing display window in seconds (default: 40 s). The underlying store retains more samples than the window shows; `MAX_SAMPLES = 4,000` acts as the absolute hard ceiling. At 100 Hz, 4,000 samples covers exactly 40 seconds — the two parameters are intentionally aligned. At lower sample rates (e.g. 50 Hz), a longer window (up to 80 s) can be entered without hitting the cap.

19.5 Filters

Three **preset filter buttons** are available alongside the raw (no filter) option:

Button	Type	Passband	Typical use
Raw	None	Full spectrum	Baseline monitoring; amplitude in counts
1–10 Hz BP	Bandpass	1.0 – 10.0 Hz	Local seismicity; removes microseisms and HF noise
5 Hz LP	Low-pass	DC – 5.0 Hz	Long-period monitoring; isolates structural modes

The active filter is highlighted with a teal background. Filters apply **zero-phase** (forward+backward Butterworth) and run on the Rust backend via `apply_filter`. The filtered output feeds the amplitude unit conversion and the STA/LTA detector simultaneously.

An additional **HP filter** toggle in the analysis controls row enables a standalone high-pass filter (default cutoff 0.1 Hz). This is applied **after** the preset filter, allowing combined bandpass + HP workflows. The cutoff frequency is editable.

19.6 Y-Axis Scaling and Display Units

Auto Y (checked by default) lets Plotly auto-range the Y-axis to fit the visible data. Unchecking it reveals **Y min** and **Y max** number inputs for fixed scaling — useful for maintaining consistent amplitude reference across multiple stations or events.

The **Unit** dropdown selects the display unit for the Y-axis and amplitude readout. Units are grouped:

Group	Units	Requires calibration?
Raw	counts, %	No — counts is always available; % uses fixed VPC factor
Acceleration	m/s², Gal, g	Yes — requires VPC and sensitivity in calibration store
Velocity	m/s, mm/s	Yes
Displacement	m, mm, µm	Yes

If a unit requires calibration data not yet loaded, the unit appears greyed out in the dropdown and a **yellow warning bar** appears below the controls explaining what calibration parameter is missing.

The Y-axis title updates automatically to reflect the selected unit (e.g. *"Acceleration (m/s²)"*).

19.7 Amplitude Measurement

1. Enable **Box Select Mode** (checkbox in the amplitude controls row). The chart cursor changes to a crosshair/selection indicator.
2. Click and drag across the time window of interest to define the measurement region.
3. On release, `plotly_selected` fires. The selected time range is mapped back to sample indices on the **Analysis stream**, and `calculate_amplitude` is invoked.

4. The result is displayed in the **Amplitude** readout badge (monospace, teal border) in the selected unit.
5. Click **Clear** to dismiss the readout without affecting picks.

The amplitude measurement is stored internally as `measuredAmplitude` and is attached automatically when saving a P-pick (§19.8).

19.8 Manual Phase Picking

The Live Waveform module supports manual P and S-wave arrival picking for event catalogue entry.

Adding picks:

- Click **"Add P-Pick"** — a red dotted vertical line appears at the centre of the current visible time window.
- Click **"Add S-Pick"** — a blue dotted vertical line appears at the same position.
- Alternatively, **click directly on the waveform** — this sets the P-pick at the clicked time.

Refining picks by dragging:

Both pick lines are fully draggable. Click and drag the red line (P) or blue line (S) to the precise arrival time. The pick time updates in real-time as you drag; `plotly_relayout` fires on drag-end and updates the internal `manualP_Pick` / `manualS_Pick` state to the new position. A toast notification confirms the updated time.

Saving picks:

1. Set the **Event ID** number input to the relevant event record in the catalogue.
2. For P-picks, set the **First Motion** dropdown: **Up (+)**, **Down (−)**, or **Unclear**. This records the polarity in the SQLite pick table.
3. Click **"Save P-Pick"** or **"Save S-Pick"** (buttons are disabled until the respective pick line exists).
4. The `save_phase_pick` Tauri command writes the pick to the `phase_picks` table with station ID, channel, phase type, timestamp (Unix seconds), measured amplitude, and polarity.
5. A toast confirms the pick ID assigned by the database.

19.9 STA/LTA Automatic Trigger

The **Enable STA/LTA** checkbox activates the short-term/long-term average trigger detector on the **Analysis stream**:

Control	Default	Description
STA (s)	0.5 s	Short-term averaging window
LTA (s)	10 s	Long-term averaging window
Threshold	3.0	Trigger fires when STA/LTA ratio ≥ this value

When enabled, `compute_sta_lta` is called on each render tick. Every sample where the ratio meets or exceeds the threshold is marked with a **solid red vertical line** overlaid on the chart. The status bar reads *"Detected N trigger breaches"*.

STA/LTA trigger lines are display-only — they do not write to the event catalogue automatically. To log a detected event, note the trigger timestamp and manually save a phase pick.

19.10 Interpreting Gaps

A **data gap** is rendered as a **break in the waveform line** — Plotly draws nothing between the last valid sample before the gap and the first sample after it, leaving a clear white discontinuity on the dark background. Gaps are inserted wherever consecutive sample timestamps differ by more than 1.5 × the expected sample interval (`1.5 × 1/sampleRate`). Near-threshold time jumps (1.05–1.5 × dt) are logged to the browser console at `WARN` level but do not break the line.

Common gap causes and diagnostic steps:

Visual	Likely cause	Action
Single narrow gap, 1–2 samples	Network packet loss	Check SeedLink latency in QC Matrix
Repeated gaps at regular intervals	Buffer overflow at the digitizer	Reduce channel count or SeedLink window
Large gap (>30 s) on all channels simultaneously	SeedLink connection drop	Reconnect via Network → SeedLink Connect
Gaps on one channel only	Single-channel hardware fault or wrong SEED ID	Inspect QC Matrix stream health for that channel
Gaps disappear after reconnect	Transient network issue	No action required

20. Spectrogram — Time-Frequency Analysis

The Spectrogram module computes and renders a **short-time Fourier transform (STFT) power spectral density** heatmap for any loaded waveform channel. It is the primary tool for discriminating event types by frequency content — local high-frequency events produce distinctly different spectral signatures from teleseisms, cultural noise, or structural resonances.

20.1 Rendering Engine

The spectrogram is rendered as a `heatmapgl` **trace** — Plotly's WebGL-accelerated heatmap type. Unlike the SVG-backed scatter traces in LiveWaveform, `heatmapgl` pushes the pixel-filling operation to the GPU, allowing large time-frequency matrices (up to `NFFT × N_windows` cells) to render at interactive speeds. The backend computation is performed in Rust via `compute_spectrogram`, which returns three arrays:

- `x` — centre time of each STFT window (Unix seconds)
- `y` — frequency bins (Hz), from 0 to Nyquist
- `z` — 2D power matrix in **dB** (`10 × log10(PSD)`)

The computation uses a **Hann window** with **75% overlap** (`noverlap = floor(nfft × 0.75)`), consistent with the Welch method used elsewhere in the DSP chain.

20.2 Controls

Control	Options	Default	Description
Stream	All loaded channel IDs	First stream	Channel to compute spectrogram for
NFFT	128, 256, 512, 1024	256	FFT window length in samples. Controls the time-frequency resolution trade-off (see §20.4)
Color map	Viridis, Plasma, Magma, Jet	Viridis	Colour palette mapping dB power to pixel colour
dB lower	Any number, step 5	−120 dB	Minimum of the colour scale — values at or below this are mapped to the cold colour
dB upper	Any number, step 5	−20 dB	Maximum of the colour scale — values at or above this are mapped to the warm colour

The spectrogram **recomputes automatically** whenever the stream selection or NFFT changes. The dB limits and colour map are applied purely to the rendered heatmap — they do not trigger a recompute.

20.3 Reading the Display

The heatmap axes are:

- **X-axis (horizontal): Time (s)** — left = earliest sample, right = most recent sample.
- **Y-axis (vertical): Frequency (Hz)** — bottom = 0 Hz (DC), top = Nyquist frequency (= sample rate / 2). For a 100 Hz channel, the Y-axis spans 0–50 Hz.
- **Colour intensity: Power (dB)** — the colour bar on the right shows the mapping. Warm colours (yellow/white in Viridis, bright in Plasma) indicate high spectral power; cold colours (dark blue/purple) indicate low power or noise floor.

The dB scale is absolute — `10 × log10(counts² / Hz)`. The absolute values depend on the sensor's full-scale counts and are not calibrated to physical units in this view. Use the **Kinematics** module for physically calibrated spectral analysis.

20.4 NFFT and the Time-Frequency Trade-off

NFFT (the FFT window length) governs the fundamental trade-off between time resolution and frequency resolution:

NFFT	At 100 Hz: window duration	Frequency resolution	Best for
128	1.28 s	0.78 Hz	Short impulsive signals (explosions, impact events)
256	2.56 s	0.39 Hz	Standard seismicity — good general-purpose choice
512	5.12 s	0.20 Hz	Structural resonance; better modal frequency separation
1024	10.24 s	0.10 Hz	Long-period teleseismic or ambient noise analysis

Rule of thumb: use the smallest NFFT that gives adequate frequency resolution for the phenomenon of interest. Large NFFT on a short record produces few time windows and a coarse temporal picture.

20.5 Identifying Event Types by Spectral Signature

Spectral pattern	Event type
Broadband energy burst (5–40 Hz), duration < 5 s, sharp onset	Local earthquake or quarry blast
Energy concentrated below 2 Hz, long duration (> 30 s), emergent onset	Teleseismic P-wave arrival
Continuous narrow horizontal band (one or two frequencies)	Structural resonance or machinery line frequency
Broadband low-level background, no sharp onset	Ambient noise / cultural noise floor
Short-duration high-frequency burst (> 20 Hz), < 1 s	Impact event, falling object, local mechanical source
Periodic energy pulses at regular intervals	Machinery (pump, compressor, traffic)

Reading a local earthquake:

1. Set NFFT = 256 (standard).
2. The P-wave arrival appears as a **vertical broadband energy stripe** at the P-arrival time, typically spanning 5–30+ Hz for a local event.
3. The S-wave arrives later as a **second, often stronger stripe** — S-waves carry more energy on horizontal components (N and E channels). Switch the **Stream** selector to the N or E channel for clearer S-wave contrast.
4. The **corner frequency** (the frequency below which the spectral level flattens to a plateau) is inversely related to the event magnitude — larger events have lower corner frequencies. The transition from the flat plateau to the roll-off at higher frequencies is visible in the spectrogram as a change from bright uniform colour to progressively cooler colour toward the top of the heatmap.
5. **Coda** (scattered energy) appears as gradually diminishing broadband energy after the S-wave stripe, tapering toward the noise floor over tens of seconds.

Separating teleseisms from local events:

Teleseismic arrivals are characterised by energy concentrated in the **0.01–1 Hz range** (below the typical spectrogram range for 100 Hz data). At 100 Hz sample rate, the spectrogram bottom ~1 Hz bins carry the teleseismic energy; a broadening of the low-frequency bins without a corresponding burst at higher frequencies indicates a distant event. Set NFFT = 1024 to resolve sub-1 Hz structure.

20.6 Adjusting the Colour Scale for Visibility

If the heatmap appears uniformly cold (all dark) or uniformly warm (saturated):

1. Note the dynamic range of the event: most of the interesting energy typically spans a 20–40 dB range.
2. Adjust **dB upper** downward until the event bursts saturate — this brightens the noise floor and makes weak arrivals visible.
3. Adjust **dB lower** upward to suppress the noise floor if it is masking the event.
4. For monitoring ambient noise baselines, **Plasma** or **Magma** colormaps often provide better contrast in the low-dB range than Viridis.

Practical starting point: set dB lower = (noise floor estimate − 5 dB), dB upper = (peak event power + 5 dB). The noise floor is visible as the stable low-level colour that persists between events.

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